ESS (EMERGENCY STOP SIGNAL SYSTEM)

id091800011500

Outline

- The emergency signal system (ESS) warns the driver of a vehicle following behind of an emergency braking situation by automatically and rapidly flashing the hazard warning lights when the vehicle decelerates suddenly while traveling at high speeds.

Function

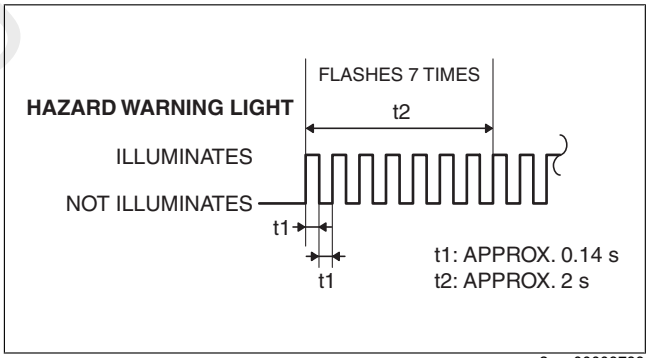
- The BCM gives priority to the emergency signal system (ESS) operation even if the turn switch signal is received during the emergency signal system (ESS) operation.
- When the BCM receives a hazard warning switch on signal during the emergency signal system (ESS) operation, it stops the emergency signal system (ESS) operation and gives priority to the hazard warning light flashing operation.
- The BCM controls the emergency signal system (ESS) based on the following signals:

Signal name	Sending module/part name	Communication method
Vehicle speed signal	PCM	HS-CAN
Brake switch condition signal		
Wheel speed (LF, RF, LR, RR) signal	DSC HU/CM	
ABS operation condition signal		
Vehicle acceleration speed signal		
Turn switch signal	Start stop unit	
Hazard warning switch signal		

- The emergency signal system (ESS) has two warning modes; the sudden deceleration hazard warning mode in which the hazard warning lights flash rapidly, and the emergency stop hazard warning mode in which the hazard warning lights flash normally.

Sudden deceleration hazard warning mode

- The sudden deceleration hazard warning mode operates when the brake pedal is depressed while driving at a vehicle speed of **60 km/h {37 mph} or more** and emergency braking is determined based on the vehicle's deceleration.
- When the sudden deceleration hazard warning mode operates the hazard warning lights flash at **3.5 Hz**.



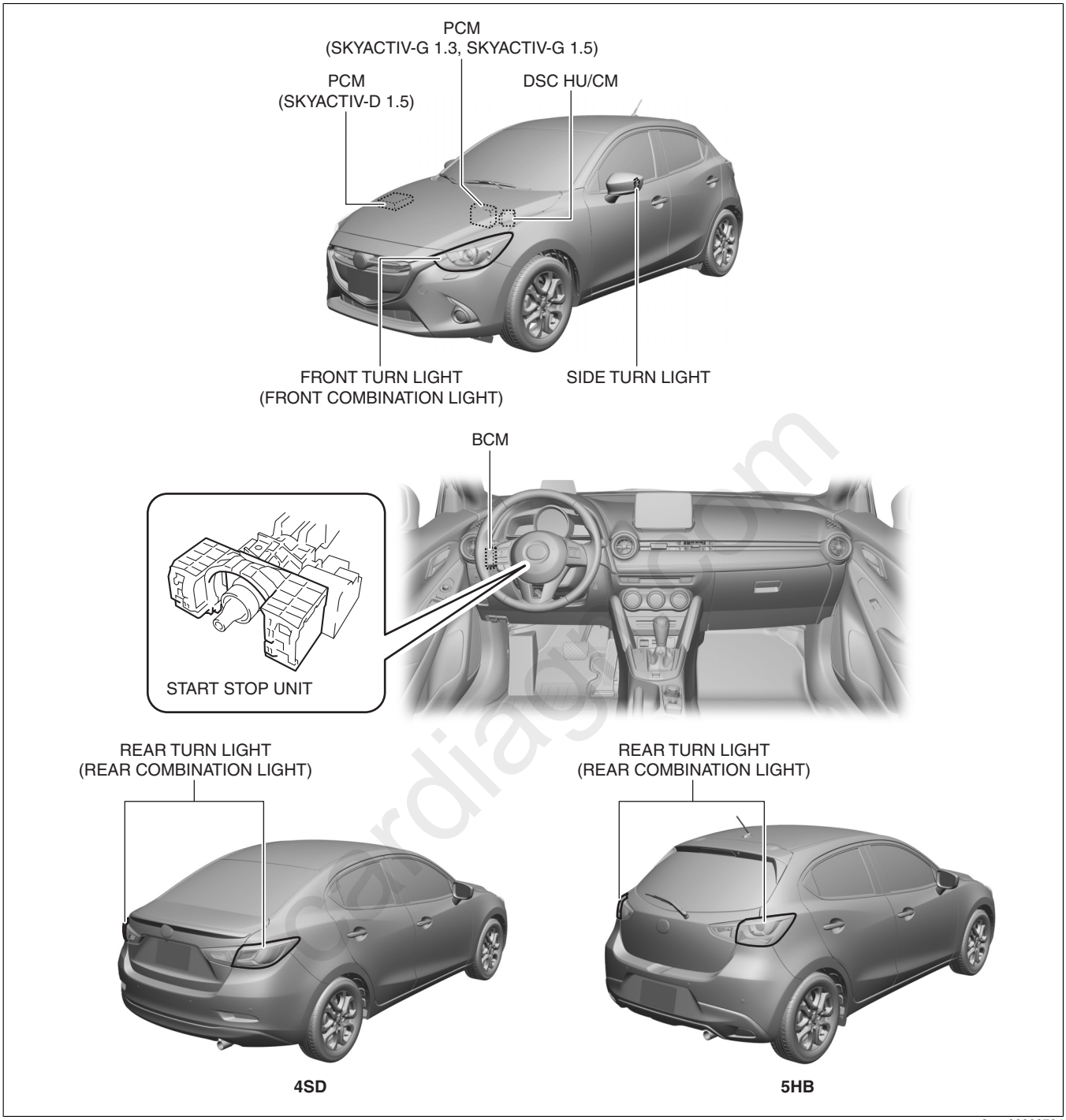
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Emergency stop hazard warning mode

- When the brake pedal is depressed at a vehicle speed of **10 km/h {6.2 mph} or less** after the sudden deceleration hazard warning mode operates, the modes switches to the emergency stop hazard warning mode.

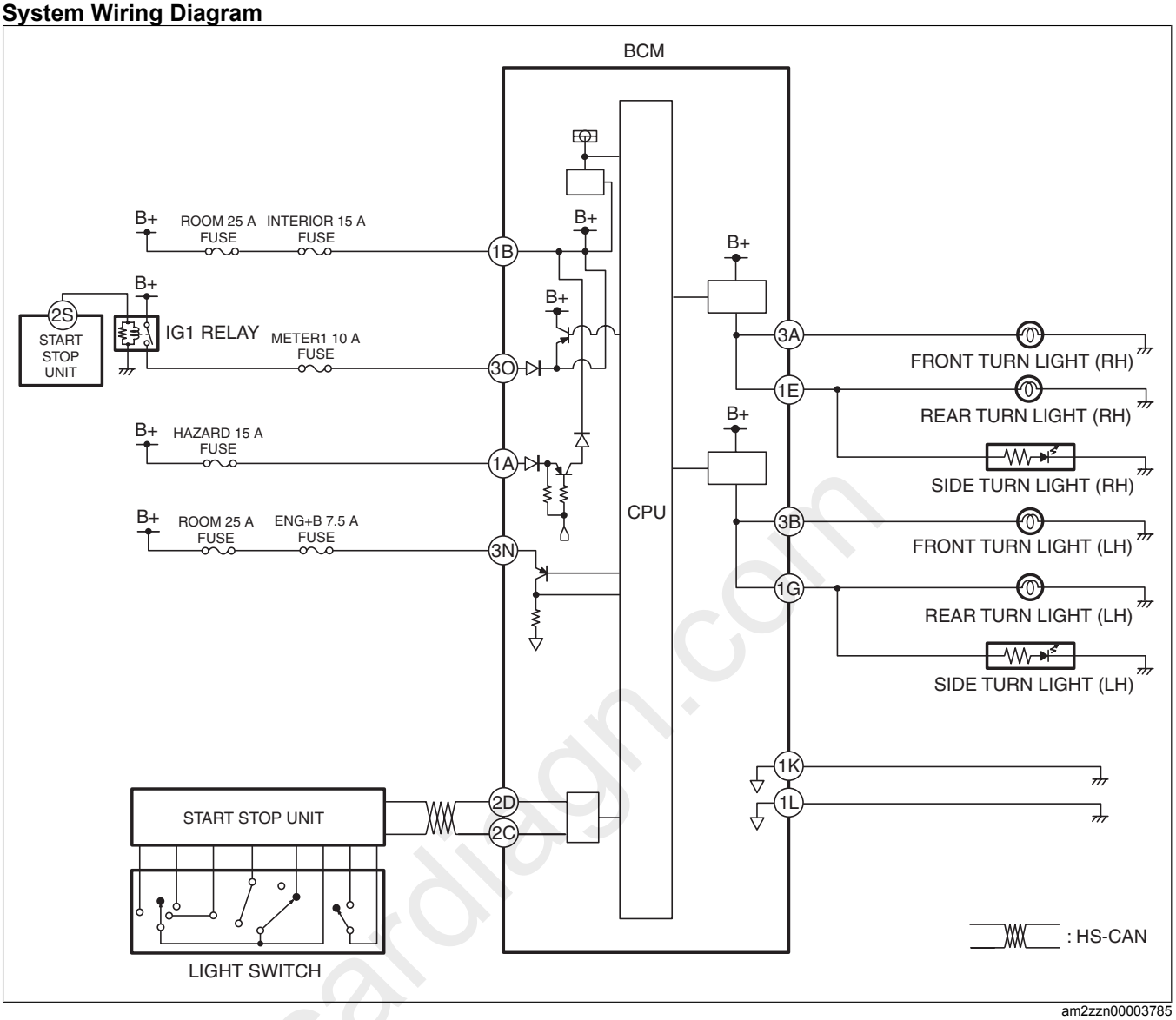
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Structural View



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System Wiring Diagram

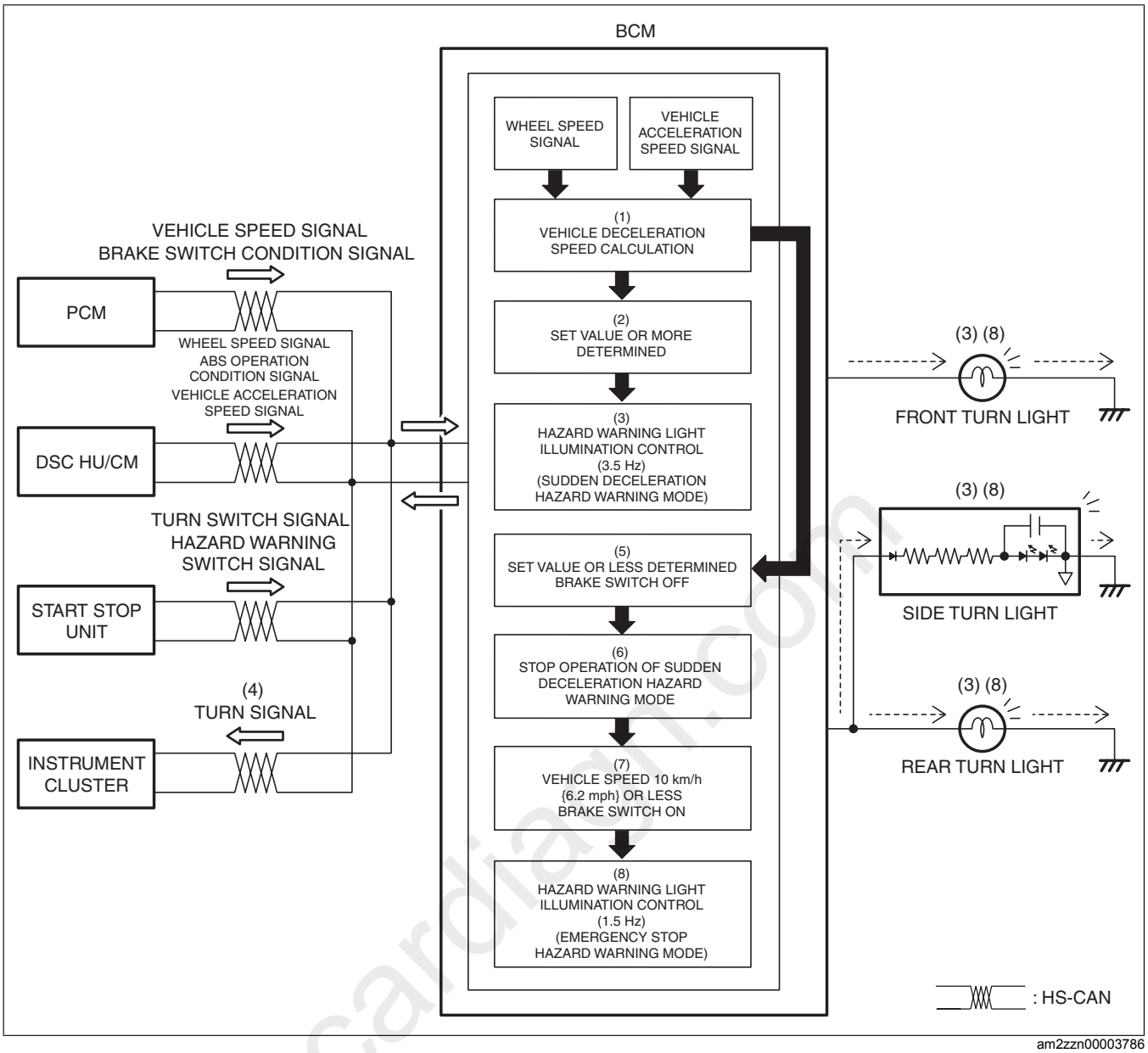


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Operation

1. If the brake pedal is depressed and the vehicle is decelerated while the vehicle is driven, the BCM calculates (1) the vehicle deceleration speed based on the wheel speed signal and the vehicle acceleration speed signal.
2. If the vehicle speed is **approx. 60 km/h {37 mph} or more** when the calculation is finished, and the calculated deceleration speed is the set value or more for the sudden deceleration hazard warning mode (2), the BCM controls (3) the flashing of the hazard warning lights at **3.5 Hz** (sudden deceleration hazard warning mode) and sends (4) a turn signal to the instrument cluster simultaneously.
3. The BCM stops (6) the operation of the sudden deceleration hazard warning mode if it receives (5) a brake switch off (brake pedal not depressed) signal or if the estimated deceleration value is the set value or less for stopping the sudden deceleration hazard warning mode.
4. When the BCM receives (7) a brake switch on (brake pedal depressed) signal at a vehicle speed of **10 km/h {6.2 mph} or less** after the sudden deceleration hazard warning mode operates, it controls (8) the flashing of the hazard warning lights at **1.5 Hz** (same as normal flashing) (emergency stop hazard warning mode).
5. When the BCM receives the brake switch off (brake pedal not depressed) signal and vehicle speed signal and determines that the vehicle has started, it stops the operation of the emergency stop hazard warning mode and turns the hazard warning lights off.



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Prevention of emergency signal system (ESS) unnecessary operation

- If the braking force inspection is performed, unnecessary operation of the emergency signal system (ESS) can be prevented if necessary.
- The operation of the emergency signal system (ESS) can be canceled by performing the procedure for preventing unnecessary operation of the brake override system. For the procedure to prevent unnecessary operation of the brake override system, refer to the [DRIVE-BY-WIRE CONTROL]/[FUEL INJECTION AMOUNT CONTROL]. (See DRIVE-BY-WIRE CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See FUEL INJECTION AMOUNT CONTROL [SKYACTIV-D 1.5].)

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